Benjamin Stahl

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EDUCATION

Lehigh University, Bethlehem, PA — *B.S. Computer Science & Business, Minor in Data Science*
GPA: 3.52 | Expected Dec 2026 | Study Abroad: Barcelona, Spain (Summer 2025)
Relevant Coursework: Computer Architecture, Systems Software, Operating Systems, Machine Learning, DS&A
UVM Training: Sunburst Design SystemVerilog UVM Verification & UVM RAL (Cliff Cummings), October 2026

WORK EXPERIENCE

Cisco, Maynard, MA | ASIC Design Verification Engineer Intern — May 2026 - August 2026

- **Verified 1.6T MACsec (IEEE 802.1AE) RX and TX datapath blocks of a coherent optical DSP ASIC**
  - Methodology used a C++ testbench with SystemVerilog coverage (VCS, Verdi, LSF)
  - Traced recurring checker failures by determining the checker was sampling slower than the DUT output, resolved missing valid values, fixed the checker, and eliminated false failures
  - Rebuilt the functional coverage model after finding false closure from missing legal lane layouts; generated bins via per-lane helper functions and a Python script enumerating all legal layouts
  - Wrote reset test incorporating soft (datapath) and hard (datapath + config); root-caused a post-reset failure to the encryption IP requiring reconfiguration, fixed it with a DPI handshake with C++ model
- **Improved verification infrastructure and nightly regression health for both blocks**
  - Fixed errors that blocked non-unity builds of the verification codebase, enabling Coverity in nightly regression; fixed 5 high-severity defects
  - Triaged ~40 nightly regression failures with a Python script that pulled seeds, configs and block owners from the results database for one-command reproduction; closed a third as stale

EarthPulse AI, Barcelona, Spain | Geospatial Data Science Intern — May 2025 - July 2025

- Automated and parallelized daily downloads of satellite imagery using Python, replacing one-at-a-time manual downloads; cut runtime 40% and had fresh data ready each morning
- Built pipelines turning 1TB+ of imagery into training pairs and triplets (multiple images of one location) for unsupervised super-resolution models

TECHNICAL SKILLS

- **Verification:** UVM (sequences, TLM, scoreboards, factory, RAL), constrained-random, functional coverage (covergroups, crosses, URG), SVA, DPI (C++↔SV)
- **EDA Tools:** Synopsys VCS, Verdi, Verilator, Slang, LSF
- **Languages:** SystemVerilog, C/C++, Python, Perl, Tcl, Bash
- **Other:** Git, SVN, Coverity, Linux, CMake/Ninja, Docker, CI/CD

TECHNICAL PROJECTS

Tclslang (Tcl Extension for Verilog Analysis) — C/C++, Slang (SystemVerilog), Tcl

- Built a Tcl extension on the slang SystemVerilog compiler for scripting design checks in an EDA-shell style, elaborating multi-file designs into an API (get_module, get_cells, get_ports, get_driver)
- Resolved full instance hierarchies through generate blocks, instance arrays and interface/modport ports, reporting each port's direction, net/variable kind, data type, width and dimensions
- Classified every port connection as net, variable, constant, expression or interface, with source text and data type, and surfaced slang diagnostics as Tcl errors

Encrypted Concurrent Key-Value Store — C++, POSIX, OpenSSL

- Built a multi-client C++ key-value server using two-phase locking, so each in-memory update and its durable log record commit atomically and stay correctly ordered under concurrent writers
- Made writes crash-resilient with direct POSIX I/O (write/fsync) and a custom binary log format, so no committed update is lost if the server crashes
- RSA handshake and AES-encrypted client–server traffic; salted SHA-256 hashes authenticate writes

LEADERSHIP & INVOLVEMENT

Computer Science & Business Association, VP of Development / Technical Development Chair (2024–2026)

- Founded the **Lehigh Valley Collegiate Hackathon** (~100 students, six universities); raised **$10K** in sponsorships

CSB Connect Pre-Orientation Program, Co-Founder & Leader (2024, 2025)

- Co-designed the program from scratch with a faculty advisor, planning the schedule and logistics, then led it for its first two years, mentoring incoming CSB first-years through their transition to Lehigh